

IAM – IDENTITY ACCESS MANAGEMENT

Class no:3

What is IAM in AWS?

- **Definition:** AWS Identity and Access Management (IAM) is a web service (security) that helps you securely control access to AWS resources.
- **Purpose:** It manages authentication (who can sign in) and authorization (what they can do).
- **Benefit:** Provides fine-grained access control, centralized management, and improved security by preventing unauthorized access. It works based on the principle of least privilege.

Four Key Factors of IAM

IAM is made up of four core blocks namely,

- IAM User
- IAM Group
- IAM Rules
- IAM Policies

1.IAM User:

Individual identities created in IAM with specific credentials (username, password, access keys).

Eg: A developer with access to EC2 instances.

Hands on : Creating an IAM user named Nithya

Step 1: Login to the Root user account.

aws

Sign In

Access your AWS account by user type.

User type [\(not sure?\)](#)

☒ **Root user**
Account owner that performs tasks requiring unrestricted access.

☐ **IAM user**
User within an account that performs daily tasks.

Email address

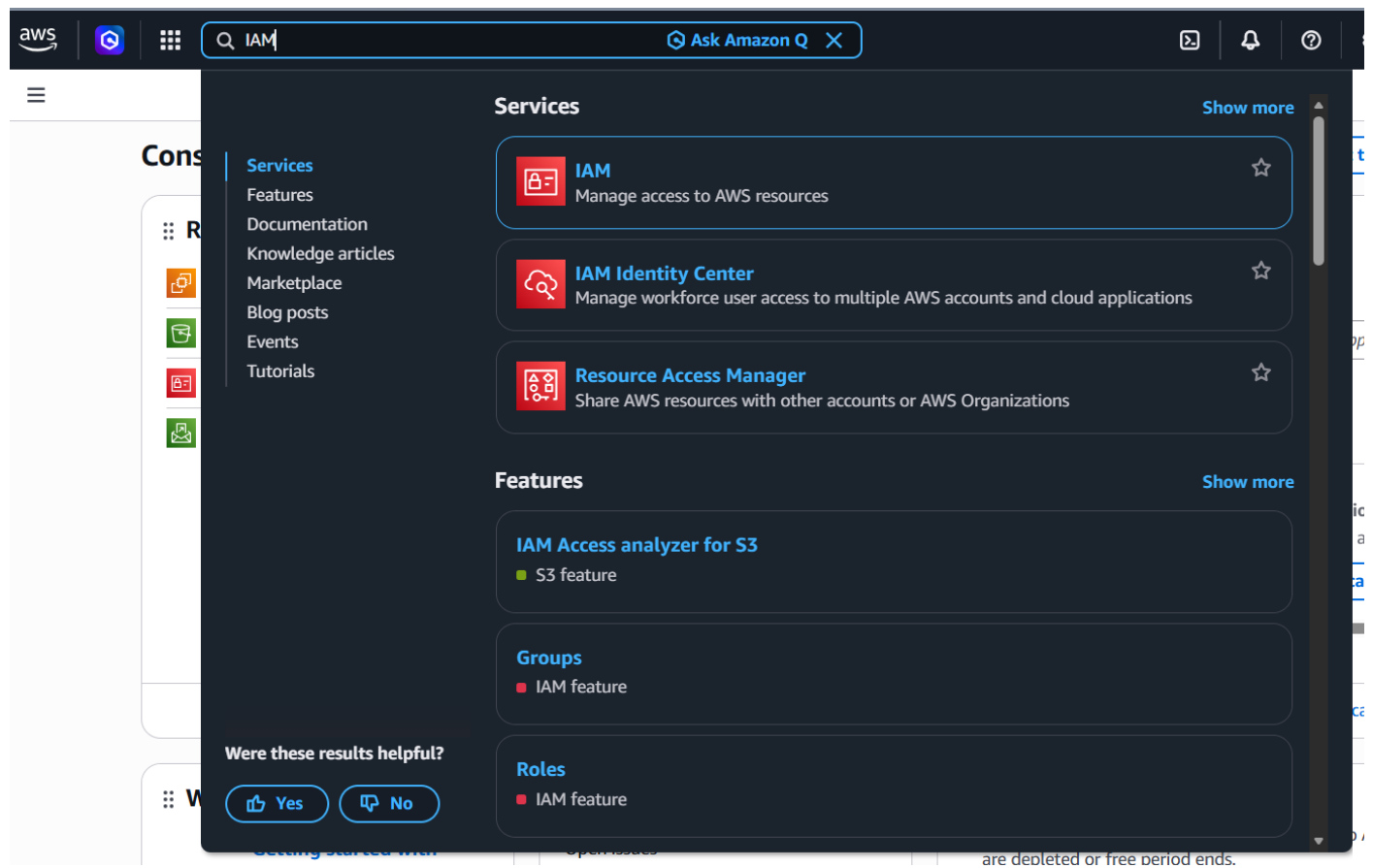
Next

OR

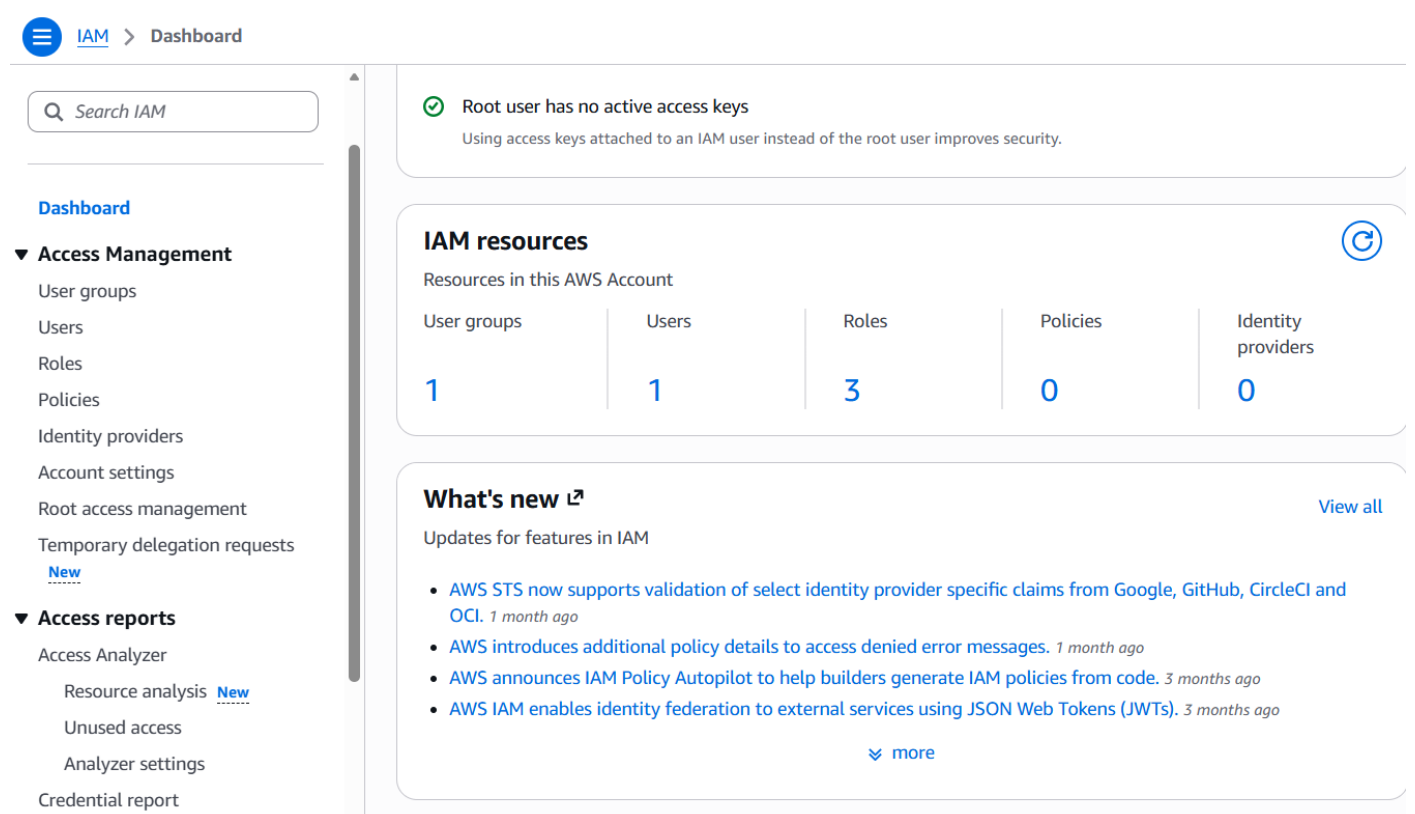
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Step 2: Search IAM



Step 3: Click & Open IAM



Step 4: To create a IAM user click the **users** option in the left side and click **create user**.

The screenshot shows the AWS IAM console 'Users' page. On the left, the 'Access Management' menu is expanded, showing 'Users' as the selected option. The main content area is titled 'Users (1)' and includes a search bar and a table of users. The 'Create user' button in the top right corner is circled in green.

☐	User name	Path	Group	Last activity	MFA	Password age	Console last sign-in	Ac
☐	varun	/	1	-	-	12 days	-	-

Step 5: Specify user details.

User name: Nithya, Password: auto-generated, enable Provide user access to the AWS Management Console, enable Users must create a new password at next sign-in and click on next.

The screenshot shows the 'Specify user details' step in the AWS IAM console. The 'User name' field is set to 'Nithya'. The 'Provide user access to the AWS Management Console' checkbox is checked. The 'Autogenerated password' option is selected for the console password. The 'Users must create a new password at next sign-in' checkbox is also checked. The 'Next' button is highlighted in orange.

User details

User name

Nithya

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and +, -, ., @, _ (hyphen)

☒ Provide user access to the AWS Management Console - optional

In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

☒ Autogenerated password

You can view the password after you create the user.

☐ Custom password

Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # \$ % ^ & * () _ + - (hyphen) = { } [] | ' "

☐ Show password

☒ Users must create a new password at next sign-in - Recommended

Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

[Learn more](#)

Cancel Next

Step 5: Set permissions, we don't add a group now, we will create a new group. So click next.

The screenshot shows the 'Set permissions' step in the AWS IAM console. The 'Add user to group' option is selected. The 'User groups' table shows one group named 'developer' with 0 users. The 'Create group' button is highlighted in orange.

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☒ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☐ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1)

Search

☐	Group name	Users	Attached policies	Created
☐	developer	0	AlexaForBusinessGatewayExecution	2026-02-27 (12 days ago)

[Create group](#)

Set permissions boundary - optional

Cancel Previous Next

Step 6: Create user with the only **policy: IAMUserchangepassword** and click create user.

The screenshot shows the 'Review and create' step in the AWS IAM console. On the left, a progress bar indicates the steps: Step 1 (Specify user details), Step 2 (Set permissions), Step 3 (Review and create), and Step 4 (Retrieve password). The main content area is titled 'Review and create' and includes a sub-header 'Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.'

The 'User details' section shows the 'User name' as 'Nithya', the 'Console password type' as 'Autogenerated', and 'Require password reset' as 'Yes'.

The 'Permissions summary' section shows a table with one entry: 'IAMUserChangePassword' (AWS managed) with the 'Used as' permissions policy.

The 'Tags - optional' section states: 'Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user. No tags associated with the resource.' There is an 'Add new tag' button and a note: 'You can add up to 50 more tags.'

At the bottom right, there are three buttons: 'Cancel', 'Previous', and 'Create user'.

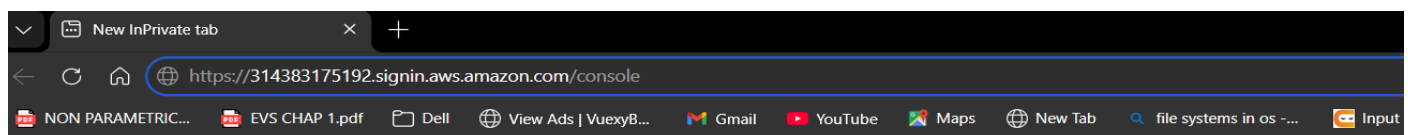
Step 7: A new IAM user will be created and the details can be downloaded as a CSV file for sharing details to the respective user.

The screenshot shows the 'Retrieve password' step in the AWS IAM console. The progress bar on the left shows Step 4 (Retrieve password) as the current step. The main content area is titled 'Retrieve password' and includes a sub-header: 'You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.'

The 'Console sign-in details' section shows the 'Console sign-in URL' as 'https://314383175192.signin.aws.amazon.com/console'. The 'User name' is 'Nithya'. The 'Console password' is masked with asterisks and has a 'Show' button next to it. There is an 'Email sign-in instructions' button in the top right.

At the bottom right, there are three buttons: 'Cancel', 'Download .csv file', and 'Return to users list'.

Step 8: To login as IAM user, open the CSV file to copy the console link and paste it in a new browser (incognito browser as we are using the same system, to avoid sign out from our current root user account.)



InPrivate browsing

Step 9: The IAM user id will be Auto-populated and enter your respective user name as well as password to login.

The screenshot shows the 'IAM user sign in' page. It has a title 'IAM user sign in' with an information icon. The form includes the following fields: 'Account ID or alias (Don't have?)' with the value '314383175192', a checkbox for 'Remember this account', 'IAM username' with the value 'Nithya', and 'Password' with masked characters and a toggle icon. There is a checkbox for 'Show Password' and a link for 'Having trouble?'. At the bottom is a 'Sign in' button.

Step 10: As we enabled the Users must create a new password at next sign-in, it will ask to change password so create a new password to confirm the password change.

Step 11: Login to the account and we can see the console of the IAM user: Nithya.

(As we didn't add any permissions, it shows access denied.)

2.IAM Group:

It is a collection of users with shared permissions. It is easy to attach and change policies in group level , instead of changing them in individual user level.

Eg: Creating a group named: developer and all the developers to the group with respective policies and permissions.

Hands on : Creating an IAM Group named developer;

Step1: Go to IAM home page , Click : User groups, then create group named:developer.

(we can add required policies.)

Identity and Access Management (IAM)

User groups (0) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

Group name Users Permissions Creation time

No resources to display

Create group

Create user group

Name the group

User group name

Enter a meaningful name to identify this group.

developer

Maximum 128 characters. Use alphanumeric and '+@-.' characters.

Add users to the group - Optional (1) Info

Step 2: Now a user group name developer will be created. We can see it doesn't have any user under it.

User groups (1) Info

A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.

Search

Group name Users Permissions Creation time

developer 0 Not defined Now

Step 3: Now click on the group name: developer, then click Add users.

developer Info

Summary

User group name: developer

Creation time: March 11, 2026, 18:43 (UTC+05:30)

ARN: arn:aws:iam::314383175192:group/developer

Users Permissions Access Advisor

Users in this group (0)

An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.

Search

User name Groups Last activity Creation time

No resources to display

Step 4: Now select User: Nithya and choose add user. Nithya will be added to the developer group.

Add users to developer Info

Other users in this account (1/1)

Search

User name Group Last activity Creation time

Nithya 0 19 minutes ago 26 minutes ago

Cancel Add users

Summary

User group name
developer

[Users \(1\)](#)

[Permissions](#)

[Access Advisor](#)

Users in this group (1)

An IAM user is an entity that you create in AWS to represent the per

☐ [User name](#)

☐ [Nithya](#)

How to change the IAM user password, if he / she forgot it?

Now you have to request the Root user.

Step 1: Open the IAM console, click on user name: Nithya (respective user name, we want to change the password) , then move to **security credentials**.

Then click Manage console access (on the right side).

Identity and Access Management (IAM)

Search IAM

Dashboard

Access Management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management
- Temporary delegation requests

Access reports

- Access Analyzer
- Resource analysis
- Unused access
- Analyzer settings
- Credential report

Nithya [Info](#)

Summary

ARN: [arn:aws:iam::314383175192:user/Nithya](#)

Created: March 11, 2026, 18:21 (UTC+05:30)

Console access: [Enabled without MFA](#)

Access key 1: [Create access key](#)

Last console sign-in: [Today](#)

Permissions | Groups (1) | Tags | **Security credentials** | Last Accessed

Console sign-in

Console sign-in link: [https://314383175192.signin.aws.amazon.com/console](#)

Console password: Updated 23 minutes ago (2026-03-11 18:33 GMT+5:30)

Last console sign-in: [28 minutes ago \(2026-03-11 18:28 GMT+5:30\)](#)

Multi-factor authentication (MFA) (0)

Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)

Remove Resync [Assign MFA](#)

Manage console

Step 2: Choose reset password, and choose autogenerate password, a new password will be created for the console, then we should login use it, followed by creating a new password of our own.

Manage console access

☐ Disable console access
Disabling removes the pre-existing password.

☒ **Reset password**
Autogenerate or set a new custom password.

Console password

☒ **Autogenerated password**

☐ Custom password

☐ User must create new password at next sign-in
Users automatically get the [IAMUserChangePassword](#) policy to allow them to change their own password.

☐ Revoke active console sessions
User automatically gets the following inline policy attached, to revoke their active sessions.

AWSRevokeOlderSessions

[Cancel](#) [Reset password](#)